Beijing Summer School

Appendix 2: Kolmogorov Forward Equation

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"The theory of probability as mathematical discipline can and should be developed from axioms in exactly the same way as Geometry and Algebra."

–Andrey Kolmogorov.

"Every mathematician believes that he is ahead of the others. The reason none state this belief in public is because they are intelligent people."

–Andrey Kolmogorov.

Goals

- Derive a law of motion for the population distribution
  (“Kolmogorov Forward Equation” or “Fokker-Planck Equation”)
- Study important example diffusions
- Learn tricks for solving Parabolic differential equations analytically
- Explore conditions for existence of a stationary distribution

References

- Introductory guides to KFEs: Pavliotis (2014), Klebaner (2012)
- Classic reference on KFEs: Riskin (1996)
- Guide to boundary conditions: Karlin and Taylor (1981), Gardiner (2004)
- Techniques for solving differential equation:
  - Ordinary differential equations: Constanda (2013)
  - Linear partial differential equations: Myint-U & Debnath(2007)

Optional Material

- Some of the material in this lecture is more “technical” than is required for an economics course
- I have indicated these sections with an asterix and put them in an appendix.

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Environment

- Population: continuum of agents $I := [0, 1]$
- Each agent, $i \in I$, has a state variable $X_t^i \in \mathbb{R}$ that follows:

$$dX_t^i = \mu_t(X_t^i)dt + \sigma_t(X_t^i)dW_t^i \tag{1}$$

  where:
  - W_t^i is an independent 1D Brownian motion processes.
  - $\mu_t(x)$, $\sigma_t(x)$ are Lipschitz in x and t
- Population notation:
  - $G_t = \mathcal{L}(X_t^i)$: population distribution across state X_t^i at time t
  - g_t : population density across state X_t at time t

Examples

- Household wealth distribution: $X_t^i = a_t^i$, where a_t^i is agent i's wealth.
- Firm productivity distribution: $X_t^i = z_t^i$, where z_t^i is firm i's productivity.
- Spatial distribution: $X_t^i = n_t^i$, where n_t^i is population in region i.

"Want" Operator

- Goal: find differential equation (DE) for evolution of population density, g_t, over time.

- This equation is called the:
  - Kolmogorov Forward Equation (KFE), or the
  - Fokker-Planck Equation (FPE)

Kolmogorov Forward Equation

Theorem

Suppose that each agent's state, X_t^i, evolves according to:

$$dX_t^i = \mu_t(X_t^i)dt + \sigma_t(X_t^i)dW_t^i.$$

Then, the population density, g_t satisfies the Kolmogorov Forward equation:

$$\partial_t g_t(x) = -\partial_x[\mu_t(x)g_t(x)] + \frac{1}{2}\partial_{xx}[\sigma_t^2(x)g_t(x)]$$

subject to an initial condition denoted by:

$$g_0(x) = \psi(x)$$

Proof: Outline

- Idea: Study dynamics of finite agent population then take limit as number of agents $\to \infty$. ("Propogation of chaos" technique.)
- Step 1: Set up the problem with a finite number of agents.
- Step 2: Apply Ito's lemma to describe evolution of "test" density, ϕ_t.
- Step 3: Take limit as $N \to \infty$.
- Step 4: Apply integration by parts to get differential equation for g_t.
- Step 5: Rearrange and conclude.

Proof: Step 1: Set-up

- Population: $N < \infty$ agents with X_t^i that follow equation (1).
- Define the finite agent (empirical) density function:

$$\hat{g}_t^N(x) := \frac{1}{N}\sum_{i=1}^{N} \delta_{X_t^i}(x)$$

  where $\delta_{X_t^i}(x)$ is the dirac-delta measure on the real line.

- The limit of the finite agent density is:

$$g_t^N \to g_t, \quad N \to \infty$$

Aside: Dirac-Delta Function

- Informally, the Dirac-Delta function, $\delta(x - x_0)$ is a function satisfying:

$$\delta(x) = \begin{cases} 0, & \text{if } x \neq x_0 \\ \infty, & \text{if } x = x_0 \end{cases} \qquad \int_{-\infty}^{\infty} \delta(x) = 1$$

- Conceptually, it is the density for a "point-mass" at x_0

Proof: Step 2: Apply Ito's Lemma

- We would like to "differentiate" the finite agent density $\hat{g}_t^N(x) := \frac{1}{N}\sum_{i=1}^N \delta_{X_t^i}(x)$.
- However, Dirac-delta functions are too difficult to work with directly.
- Instead, we differentiate $\frac{1}{N}\sum_{i=1}^N \phi_t(X_t^i)$, for arbitrary "test function" ϕ_t satisfying:
  - $\phi_t : \mathbb{R} \to \mathbb{R}$
  - ϕ_t is smooth (infinitely differentiable)
  - ϕ_t has compact support (is zero outside some bounded interval)

Proof: Step 2: Apply Ito's Lemma

- Now, by applying Ito's lemma to $\phi_t(X_t^i)$, we have that:

$$\frac{1}{N}\sum_{i=1}^{N}\phi_t(X_t^i) = \frac{1}{N}\sum_{i=1}^{N}\left(\phi_0(X_0^i) + \int_0^t \partial_s\phi_s(X_s^i)ds + \int_0^t \partial_x\phi_s(X_s^i)dX_s^i + \frac{1}{2}\int_0^t \partial_{xx}\phi_s(X_s^i)d[X^i, X^i]_s\right)$$

- Which, after rearranging, becomes:

$$\begin{aligned}
&\frac{1}{N}\sum_{i=1}^{N}\phi_t(X_t^i) - \frac{1}{N}\sum_{i=1}^{N}\phi_0(X_0^i) \\
&\quad = +\frac{1}{N}\sum_{i=1}^{N}\int_0^t \Big(\partial_s\phi_s(X_s^i) + \mu_s^i(X_s^i)\partial_x\phi_s(X_s^i) + \frac{1}{2}\sigma_s^2(X_s^i)\partial_{xx}\phi_s(X_s^i)\Big)ds \\
&\quad\quad + \frac{1}{N}\sum_{i=1}^{N}\int_0^t \sigma_s(X_s^i)\partial_x\phi_s(X_s^i)dW_s^i
\end{aligned} \tag{2}$$

Proof: Step 3: Take Limit $N \to \infty$

- By the Law of Large Numbers (LLN), we have that:

$$\lim_{N\to\infty} \frac{1}{N} \sum_{i=1}^{N} \phi_t(X_t^i) = \mathbb{E}\left[\phi_t(X_t)\right] = \int_{-\infty}^{+\infty} \phi_t(x) g_t(x) dx$$

$$\begin{aligned}
&\lim_{N\to\infty} \frac{1}{N} \sum_{i=1}^{N} \left(\int_0^t \left(\partial_s \phi_s(X_s^i) + \mu_s(X_s^i) \partial_x \phi_s(X_s^i) + \frac{1}{2} \sigma_s^2(X_s^i) \partial_{xx} \phi_s(X_s^i) \right) ds \right) \\
&= \mathbb{E}\left[\int_0^t \left(\partial_s \phi_s(X_s^i) + \mu_s(X_s^i) \partial_x \phi_s(X_s^i) + \frac{1}{2} \sigma_s^2(X_s^i) \partial_{xx} \phi_s(X_s^i) \right) ds \right] \\
&= \int_{-\infty}^{\infty} \int_0^t \left(\partial_s \phi_s(x) + \mu_s(x) \partial_x \phi_s(x) + \frac{1}{2} \sigma_s^2(X_s^i) \partial_{xx} \phi_s(x) \right) g_s(x) ds dx
\end{aligned}$$

$$\lim_{N\to\infty} \frac{1}{N} \sum_{i=1}^{N} \int_0^t \sigma_s(X_s^i) \partial_x \phi_s(X_s^i) dW_s^i = \mathbb{E}\left[\int_0^t \sigma_s(X_s^i) \partial_x \phi_s(X_s^i) dW_s^i \right] = 0$$

Proof: Step 3: Take Limit $N \to \infty$

- And so (after rearranging) the limit of equation (2) is:

$$\begin{aligned}\int_{-\infty}^{+\infty} \left(\phi_t(x)g_t(x) - \phi_0(x)g_0(x)\right) dx \\ = +\int_{-\infty}^{+\infty}\int_0^t \partial_s\phi_s(x)g_s(x)dsdx + \int_{-\infty}^{\infty}\int_0^t \mu_s(x)\partial_x\phi_s(x)g_s(x)dsdx \\ + \int_{-\infty}^{\infty}\int_0^t \frac{1}{2}\sigma_s^2(x)\partial_{xx}\phi_s(x)g_s(x)dsdx\end{aligned}$$

Proof: Step 4: Apply Integration by Parts

- Use integration by parts to exchange derivatives of ϕ for derivatives of g:
- E.g. Integrating the first term on the RHS:

$$\begin{aligned}
&\int_{-\infty}^{+\infty}\int_0^t \partial_s\phi_s(x)g_s(x)dsdx \\
&= \int_{-\infty}^{+\infty}\left(\left[\phi_s(x)g_s(x)\right]_0^t - \int_0^t \phi_s(x)\partial_s g_s(x)ds\right)dx \\
&= \int_{-\infty}^{+\infty}\left(\phi_t(x)g_t(x) - \phi_0(x)g_0(x) - \int_0^t \phi_s(x)\partial_s g_s(x)ds\right)dx
\end{aligned}$$

- E.g. Integrating the second term on the RHS:

$$\begin{aligned}
&\int_{-\infty}^{\infty}\int_0^t \mu_s(x)\partial_x\phi_s(x)g_s(x)dsdx \\
&= \int_0^t\left(\left[\phi_s(x)\mu_s(x)g_s(x)\right]_{-\infty}^{+\infty} - \int_{-\infty}^{\infty}\phi_s(x)\partial_x[\mu_s(x)g_s(x)]dx\right)ds \\
&= -\int_0^t\int_{-\infty}^{\infty}\phi_s(x)\partial_x[\mu_s(x)g_s(x)]dxds
\end{aligned}$$

Proof: Step 4: Apply Integration by Parts

- So, after applying integration by parts on each term:

$$
\begin{aligned}
&\int_{-\infty}^{+\infty} \left(\phi_t(x)g_t(x) - \phi_0(x)g_0(x)\right) dx \\
&= \int_{-\infty}^{+\infty} \left(\phi_t(x)g_t(x) - \phi_0(x)g_0(x) - \int_0^t \phi_s(x)\partial_s g_s(x)ds\right) dx \\
&- \int_0^t \int_{-\infty}^{\infty} \phi_s(x)\partial_x[\mu_s(x)g_s(x)]dxds + \frac{1}{2}\int_0^t \int_{-\infty}^{+\infty} \phi_s(x)\partial_{xx}[\sigma_s^2(x)g_s(x)]dxds
\end{aligned}
$$

- Which gives:

$$
0 = \int_{-\infty}^{+\infty} \int_0^t \left(- \partial_s g_s(x) - \partial_x[\mu_s(x)g_s(x)] + \frac{1}{2}\partial_{xx}[\sigma_s^2(x)g_s(x)]\right)\phi_s(x)dsdx
$$

Proof: Step 5: Differentiate and Conclude

- Differentiate with respect to t

$$0 = \int_{-\infty}^{+\infty} \Big(- \partial_t g_t(x) - \partial_x[\mu_t(x) g_t(x)] + \frac{1}{2}\partial_{xx}[\sigma_t^2(x) g_t(x)] \Big) \phi_t(x) dx$$

- Since this equation is true for all test function ϕ_t:

$$\begin{aligned} 0 =& - \partial_t g_t(x) - \partial_x[\mu_t(x) g_t(x)] + \frac{1}{2}\partial_{xx}[\sigma_t^2(x) g_t(x)] \\ \Rightarrow \partial_t g_t(x) =& - \partial_x[\mu_t(x) g_t(x)] + \frac{1}{2}\partial_{xx}[\sigma_t^2(x) g_t(x)] \quad \square. \end{aligned}$$

Alternative Derivation Approach

- Two ways to think about the density, $g_t(x)$:
  1. Cross Sectional Density:
     - Continuum of agents with states, X_t^i, that evolves by:

$$dX_t^i = \mu_t(X_t^i)dt + \sigma_t(X_t^i)dW_t^i, \quad X_0^i = y^i$$

     - $g_t(x)$ is the population density across states, X_t^i, at time t
  2. Transition Density:
     - Single agent with state, X_t, that evolves by:

$$dX_t = \mu_t(X_t)dt + \sigma_t(X_t)dW_t, \quad X_0 = y$$

     - $P_t(y,x) := \mathbb{P}(X_t \leq x | X_0 = y)$: transition distribution
     - $g_t(x) := \frac{d}{dx} P_t(y,x)$: transition density

- Can also use second definition to derive KFE.

Interpretation

- Integrate KFE with respect to x:

$$\partial_t G_t(x) = -\mu_t(x)g_t(x) + \frac{1}{2}\partial_x[\sigma_t^2(x)g_t(x)]$$

- Describes how $G_t(x) = \mathbb{P}(X_t \leq x)$ is changing over time at x:
  - First term:
    - If $\mu_t(x) > 0$, then $X_t \uparrow$ and so mass below x is $\downarrow$
    - If $\mu_t(x) < 0$, then $X_t \downarrow$ and so mass below x is $\uparrow$
  - Second term:
    - At any x, the random noise dW_t spreads out the population,
    - If $\partial_x[\sigma_t^2(x)g_t(x)] > 0$, then more mass and noise to the right of x and so the net impact of noise increases the mass below x, and
    - If $\partial_x[\sigma_t^2(x)g_t(x)] < 0$, then more mass and noise to the left of x and so the net impact of noise decreases the mass below x

Flux and Conversation of Probability Mass

- Can rewrite the KFE using the notation:

$$\partial_t g_t(x) + \partial_x J_t(x) = 0 \quad \Leftrightarrow \quad \partial_t G_t(x) = -J_t(x)$$

where $J_t(x)$ is defined to be:

$$J_t(x) := \mu_t(x) g_t(x) - \frac{1}{2}\partial_x[\sigma_t^2(x) g_t(x)]$$

  - From the previous slide, know that $J_t(x)$ is the probability per unit of time that a particle passes through point x from left to right (i.e. the net flow of probability mass from left to right at x)
  - $J_t(x)$ is referred to as the **flux** or **probability current** at x

Flux and Conversation of Probability Mass

- Integrating the KFE w.r.t. x over an arbitrary interval (a, b):

$$\partial_t \int_a^b g_t(x)dx = \big[- J_t(x) \big]_a^b = J_t(a) - J_t(b)$$

  - I.e. the change in probability mass across the interval (a, b) corresponds to the net probability flux across the boundaries.

- Integrating the KFE w.r.t. x over the real line:

$$\partial_t \int_{-\infty}^{+\infty} g_t(x)dx = \big[- J_t(x) \big]_{-\infty}^{+\infty}$$

  - So, the KFE is a **continuity equation**: rate of probability mass accumulation within the system equals net flux across the system.
  - If no birth or death, then net flux across population is zero and:

$$\partial_t \int_{-\infty}^{+\infty} g_t(x)dx = 0 \Rightarrow \int_{-\infty}^{+\infty} g_t(x)dx = \int_{-\infty}^{+\infty} g_0(x)dx = 1$$

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Boundary Conditions

- Many applications study diffusion processes on bounded domains

- In such cases, the KFE, as stated so far, does not admit a unique solution because we have not stated the boundary conditions

- Now discuss common boundaries conditions:
  1. Absorbing boundaries,
  2. Relecting boundaries, and

- Further reading:
  - Further discussion on boundary conditions, see Gardiner (2004), and
  - Formal classification of the boundary conditions, see chapter 15 in Taylor (1981).

Boundary Notation

- Let $\mathcal{X} \subset \mathbb{R}^n$ denote the space in which the agent state variables reside. That is, $x \in \mathcal{X}$.
- Let $\partial\mathcal{X}$ denote the boundary of $\mathcal{X}$. Throughout this section, I assume that $\partial\mathcal{X}$ is smooth.
- For the univariate case in which $x \in [\underline{x}, \overline{x}]$, we have that:
  - $\mathcal{X} = [\underline{x}, \overline{x}]$, and
  - $\partial\mathcal{X} = \{\underline{x}, \overline{x}\}$.

1. Absorbing Boundaries

- Informally: say that the diffusion process, X, has an absorbing boundary if X is removed from the system once it the boundary is reached.

- Formally: this is stated as:

$$g_t(x) = 0, \quad \forall x \in \partial \mathcal{X}$$

- 1D diffusion on the interval $[\underline{x}, \overline{x}]$ with absorbing boundaries: condition becomes:

$$g_t(\underline{x}) = 0, \quad g_t(\overline{x}) = 0$$

- (Could also record the accumulation of probability at the boundary.)

2. Reflecting Boundaries

- Informally: X, has a reflecting boundary if, when X hits the boundary, it heads back into the interior of the space.

- Formally: this is stated as:

$$n \cdot J_t(x) = 0, \quad \forall x \in \partial \mathcal{X}$$

  where
  - n is the normal vector to the boundary (pointing outward),
  - $J_t(x)$ is the flux, and
  - $n \cdot J_t(x)$ represents the flow across the boundary at point x in the direction of n (i.e. in the direction out of the $\mathcal{X}$). So, the condition says that there is no flow out of the set $\mathcal{X}$ on the boundary.

- 1D diffusion on the interval $[\underline{x}, \overline{x}]$ with reflecting boundaries: condition becomes (since $n = -1$ at $\underline{x}$ and $n = 1$ at $\overline{x}$.):

$$J_t(\underline{x}) = 0, \quad J_t(\overline{x}) = 0$$

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Running Illustrative Example 1

- Continuum of households indexed by $i \in [0, 1]$.
- Each household has individual wealth $x_t^i = [a_t^i]$ that evolves by:

$$da_t^i = \underbrace{(ra_t^i - c(a_t^i))}_{\mu_a(a_t^i)} dt + \sigma dW_t^i$$

  - r is the return on savings, and
  - $c(a_t^i)$ is consumption as a function of wealth.

- Let $g_t(a)$ be a density over household wealth. Then, the KFE is:

$$dg_t(a) = \left(-\partial_a [\mu_a(a) g_t(a)] + \frac{1}{2}\sigma^2 \partial_{aa}[g_t(a)] \right) dt$$

Example 2

- Same as Example 1 but now households leave the economy when $a > \overline{a}$ or $a < \underline{a}$.

- Let $g_t(a)$ be a density over household wealth. Then, the KFE is:

$$dg_t(a) = \left(-\partial_a[\mu_a(a)g_t(a)] + \frac{1}{2}\sigma^2\partial_{aa}[g_t(a)]\right)dt$$

$$s.t. \quad g(\underline{a}) = g(\overline{a}) = 0$$

Example 3

- Same as Example 1 but now households get wealth to ensure that $a < \overline{a}$ and have wealth taken to ensure that $a < \underline{a}$.

- Let $g_t(a)$ be a density over household wealth. Then, the KFE is:

$$dg_t(a) = \left(-\partial_a[\mu_a(a)g_t(a)] + \frac{1}{2}\sigma^2 \partial_{aa}[g_t(a)] \right) dt$$
$$s.t. \quad J_t(\underline{a}) = -\mu_a(\underline{a})g_t(\underline{a}) + \frac{1}{2}\sigma^2 \partial_a[g_t(\underline{a})] = 0$$
$$s.t. \quad J_t(\overline{a}) = -\mu_a(\overline{a})g_t(\overline{a}) + \frac{1}{2}\sigma^2 \partial_a[g_t(\overline{a})] = 0$$

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Environment

- Population: continuum of agents $I := [0, 1]$
- Each agent, $i \in I$, has a state variable $X_t^i \in \mathbb{R}$ that follows:

$$dX_t^i = \mu_t(X_t^i)dt + \sigma_t(X_t^i)dW_t^i + \sigma_t^0(X_t^i)dW_t^0$$

  - W_t^i is an independent 1D Brownian motion processes.
  - W_t^0 is an common 1D Brownian motion processes.
  - $\mu_t(x)$, $\sigma_t(x)$ are Lipschitz in x and t
- Population notation:
  - $G_t = \mathcal{L}(X_t^i | \mathcal{F}_t^0)$: distribution across X_t^i at t given common shock history $\mathcal{F}_t^0$
    (Formally, common shock history is $\mathcal{F}_t^0 = \sigma(\{W_s^0 : s \leq t\})$)
  - g_t : population density across state X_t at time t given $\mathcal{F}_t^0$

Kolmogorov Forward Equation

Theorem

Suppose that each agent's state, X_t^i, evolves according to:

$$dX_t^i = \mu_t(X_t^i)dt + \sigma_t(X_t^i)dW_t^i + \sigma_t^0(X_t^i)dW_t^0.$$

Then, the population density, g_t satisfies the Kolmogorov Forward equation:

$$dg_t(x) = -\partial_x[\mu_t(x)g_t(x)]dt - \partial_x[\sigma_t^0(x)g_t(x)]dW_t^0 + \frac{1}{2}\sigma_{xx}[(\sigma_t^2(x) + (\sigma_t^0)^2(x))g_t(x)]dt$$

subject to an initial condition denoted by:

$$g_0(x) = \psi(x).$$

Proof.

Exercise. Apply the same steps as the case with only idiosyncratic risk but observe that the LLN does not apply to W_t^0. □

Example 4

- Same as Example 1 (i.e. no boundaries) but now there are common shocks to returns:

$$da_t^i = \underbrace{(ra_t^i - c(a_t^i))}_{\mu_a(a_t^i)} dt + \sigma dW_t^i + \sigma^0 dW_t^0$$

- Let $g_t(a)$ be a density over household wealth. Then, the KFE is:

$$dg_t(a) = \left(-\partial_a[\mu_a(a)g_t(a)] + \frac{1}{2}\sigma^2\partial_{aa}[g_t(a)]\right) - \partial_a[(\sigma^0)^2 g_t(a)]dW_t^0$$

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Environment

- Population: continuum of agents $I := [0, 1]$
- Each agent, $i \in I$, has a state variable $X_t^i \in \mathbb{R}$ that follows:

$$dX_t^i = \mu_t(X_t^i)dt + h_t(X_t^i)dN_t^i + h_t^0(X_t^i)dN_t^0$$

  - N_t^i is an independent 1D Poisson processes with rate λ.
  - N_t^0 is an common 1D Poisson processes with rate λ^0.
  - $\mu_t(x)$, $h_t(x)$ and $h_t^0(x)$ are Lipschitz in x and t
- Population notation:
  - $G_t = \mathcal{L}(X_t^i|\mathcal{F}_t^0)$: distribution across state X_t^i at time t given history of common shocks (Formally, common shock history is $\mathcal{F}_t^0 = \sigma(\{N_s^0 : s \leq t\})$)
  - g_t : population density across state X_t at time t given $\mathcal{F}_t^0$

Kolmogorov Forward Equation (Dist.)

Theorem

Suppose that each agent's state, X_t^i, evolves according to:

$$dX_t^i = \mu_t(X_t^i)dt + h_t(X_t^i)dN_t^i + h_t^0(X_t^i)dN_t^0$$

Then, the population distribution, G_t satisfies the KFE:

$$\begin{aligned} dG_t(x) = & -\mu_t(x)g_t(x)dt \\ & + \lambda \underbrace{\left(\mathbb{P}(X_t + h(X_t) \leq x) - \mathbb{P}(X_t \leq x)\right)}_{\text{Change in mass below } x \text{ after } dN_t^i, \forall i} dt + \underbrace{\left(\mathbb{P}(X_t + h^0(X_t) \leq x) - \mathbb{P}(X_t \leq x)\right)}_{\text{Change in mass below } x \text{ after } dN_t^0} dN_t^0 \\ = & -\mu_t(x)g_t(x)dt + \lambda\left(G_t(x - \gamma(x)) - G_t(x)\right)dt + \left(G_t(x - \gamma^0(x)) - G_t(x)\right)dN_t^0 \end{aligned}$$

where γ and γ^0 are defined by the relationships:

$$\begin{aligned} X_t + h(X_t) = x \quad &\Leftrightarrow \quad X_t = x - \gamma(x) \\ X_t + h^0(X_t) = x \quad &\Leftrightarrow \quad X_t = x - \gamma^0(x) \end{aligned}$$

Kolmogorov Forward Equation (Density)

Theorem

Suppose that each agent's state, X_t^i, evolves according to:

$$dX_t^i = \mu_t(X_t^i)dt + h_t(X_t^i)dN_t^i + h_t^0(X_t^i)dN_t^0$$

Then, the population distribution, g_t satisfies the KFE:

$$\begin{aligned} dg_t(x) = & -\partial_x[\mu_t(x)g_t(x)]dt \\ & + \lambda\left((1-\gamma'(x))g_t(x-\gamma(x)) - g_t(x)\right)dt \\ & + \big((1-(\gamma^0)'(x))g_t(x-\gamma^0(x)) - g_t(x)\big)\,dN_t^0 \end{aligned}$$

where γ and γ^0 are defined by the relationships:

$$\begin{aligned} X_t + h(X_t) = x \quad &\Leftrightarrow \quad X_t = x - \gamma(x) \\ X_t + h^0(X_t) = x \quad &\Leftrightarrow \quad X_t = x - \gamma^0(x) \end{aligned}$$

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Death and Rebirth

- Population: continuum of agents $I := [0, 1]$
- Each agent, $i \in I$, has a state variable $X_t^i \in \mathcal{X}$ that follows:

$$dX_t^i = \mu_t(X_t^i)dt + \sigma_t(X_t^i)dW_t^i$$

  - W_t^i is an independent 1D Brownian motion processes.
  - $\mu_t(x)$, $\sigma_t(x)$ are Lipschitz in x and t
- Birth and death:
  - Agents killed according to Poisson process with intensity $\lambda_k(X_t)$.
  - Agents reborn at x according to Poisson process with intensity $\lambda_b(x)$.

Continuous Distribution of Inflows and Outflows

- Suppose $\lambda_k(x)$ and $\lambda_b(x)$ are continuously distributed over $\mathcal{X}$
- KFE becomes:

$$\partial_t g_t(x) = +\underbrace{\lambda_b(x)}_{\text{Inflow}} - \underbrace{\lambda_k(x) g_t(x)}_{\text{Outflow}} - \partial_x[\mu_t(x) g_t(x)] + \frac{1}{2}\sigma_{xx}[\sigma_t^2(x) g_t(x)]$$

Non-Continuous Distribution of Inflows and Outflows

- Suppose $\lambda_k(x)$ is continuously distributed over $\mathcal{X}$
- Suppose that all agents are born at $\bar{x}$.
- For $x \in \mathcal{X} \backslash \{\bar{x}\}$, the conditional density satisfies

$$\partial_t g_t(x) = -\lambda_k(x) g_t(x) - \partial_x[\mu_t(x) g_t(x)] + \frac{1}{2}\sigma_{xx}[\sigma_t^2(x) g_t(x)]$$

  The function g_t is continuous at $\bar{x}$ but not smooth (so best avoided).

Example 5

- Same as Example 2 (i.e. absorbing boundaries but no aggregate shocks):

$$da_t^i = \underbrace{\left(ra_t^i - c(a_t^i)\right)}_{\mu_a(a_t^i)} dt + \sigma dW_t^i$$

- But now suppose that new agents arrive at rate $\lambda_e(a)$ to $a \in [\underline{a}, \overline{a}]$.

- Let $g_t(a)$ be a density over household wealth. Then, the KFE is:

$$dg_t(a) = \left(\lambda_e(a) - \partial_a[\mu_a(a)g_t(a)] + \frac{1}{2}\sigma^2\partial_{aa}[g_t(a)]\right) dt$$

Next Class

1. We study different techniques for solving KFEs.

2. We will consider finite difference, spectral methods, and neural networks.

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Higher Dimensional Processes

- Now, suppose that $X_t^i \in \mathbb{R}^n$ and follows the Itô process:

$$dX_t^i = \mu_t^i(X_t^i)dt + \sigma_t^i(X_t^i)dW_t$$

  where

  - W_t is an m dimensional Brownian motion,
  - $\mu^i(x)$ is n dimensional, and
  - $\sigma_t^i(x)$ is $n \times m$ dimensional.

- Define:

  - $\Sigma_t(x) := \sigma_t(x)\sigma_t(x)^T$
  - $\Sigma_{t,jk}$ to refer to the j, k element of Σ_t

KFE for Higher Dimensional Processes

Theorem

Suppose that each agent's state, $X_t^i \in \mathbb{R}^n$, evolves according to:

$$dX_t^i = \mu_t^i(X_t^i)dt + \sigma_t^i(X_t^i)dW_t$$

Then, the population density, g_t, satisfies the KFE:

$$\partial_t g_t(x) = -\sum_{j=1}^{n} \partial_{x_j}(\mu_{t,j}(x)g(x)) + \frac{1}{2}\sum_{j,k=1}^{n} \partial_{x_j,x_k}(\Sigma_{t,jk}(x)g(x))$$

subject to the initial condition:

$$g_0(x) = \psi(x)$$

Proof.

Can proove this using similar steps to the univariate case. However, you will need to undertake integrate parts in higher dimensions. □

Alternative Notation

- The multivariate KFE is often written using the notation:

$$\partial_t g_t(x) = -\operatorname{div}\left[\mu_t(x) g_t(x)\right] + \frac{1}{2}\operatorname{tr}\left\{D_x^2\Big(\Sigma_t(x) g_t(x)\Big)\right\}$$

  where div $[\cdot]$ is the divergence and tr $(\cdot)$ is the trace.

- This is sometimes futher abbreviated to:

$$\partial_t g_t = \nabla \cdot \Big(-\mu_t(x) g_t + \frac{1}{2}\nabla \cdot (\Sigma_t(x) g)\Big)$$

  where ∇ is the gradient vector.

Multivariate Probability Flux (or Current)

- The multivariate **probability flux (or current)** can then be defined as a n-dimensional vector, $J_t(x)$, with j element:

$$J_{t,j}(x) := \mu_{t,j}(x)g_t(x) - \sum_{j=1}^{n} \partial_{x_j}\left[\frac{1}{2}\sigma_{t,j}(x)g_t(x)\right]$$
$$= \mu_t(x)g_t - \frac{1}{2}\nabla \cdot (\Sigma_t(x)g)$$

- So, the KFE can once again be written as:

$$\partial_t g_t + \nabla \cdot J = 0.$$

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Assumptions

- Consider time invariant $\mu_t = \mu(X_t)$ and $\Sigma_t = \Sigma(X_t)$
- **A1:** Assume that $\Sigma = \sigma\sigma^T$ satisfies the **uniform ellipticity** condition: there exists $\alpha > 0$ s.t. uniformly in x:

$$\sum_{j,k=1}^{n} \Sigma_{jk}\xi_j\xi_k \geq \alpha\|\xi\|^2, \quad \forall \xi \in \mathbb{R}^n$$

- **A2:** Assume that the coefficients are smooth and satisfy the growth conditions:

$$\|\Sigma(x)\| \leq M, \|\tilde{a}(x)\| \leq M(1 + \|x\|), \|\tilde{b}(x)\| \leq M(1 + \|x\|)$$

where $\tilde{a}$ and $\tilde{b}$ are defined by:

$$\tilde{a}_j(x) = -\mu_j(x) + \sum_{j=1}^{n} \partial_{x_j}\Sigma_{jk}(x)$$

$$\tilde{b}_j(x) = \frac{1}{2}\sum_{j,k=1}^{n} \partial_{x_j,x_k}\Sigma_{jk}(x) - \sum_{j=1}^{n} \partial_{x_j}\mu_j(x)$$

Classical Solution

Definition

We say that a solution to the initial value problem for the KFE is a **classical solution** if:

1. $g_t(x) = g(x,t) \in C^{2,1}(\mathbb{R}^d, \mathbb{R}^+)$
2. $\forall T > 0$, there exists a $c > 0$ such that:

$$\|g(x,t)\|_{L^\infty(0,T)} \leq ce^{\alpha \|x\|^x}$$

3. $\lim_{t\to 0} g(x,t) = \psi(x)$

Uniqueness and Existence

Theorem

*Assume that conditions **A1** and **A2** are satisfied, and assume that* $|\psi(x)| \leq ce^{\alpha\|x\|^2}$.

- *Then, there exists a unique classical solution to the Cauchy problem for the KFE.*
- *Furthermore, there exist positive constants* K, δ *such that:*

$$|g_t|, \|D^2 g_t\| \leq K t^{(-n+2)/2} e^{-\frac{1}{2t}\delta\|x\|^2}$$

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Aside: Solving KFE With Fourier Transforms*

- The *Fourier Transform* of the density is given by:

$$\hat{g}_t(k) := \hat{F}(g_t(x)) := \int_{-\infty}^{\infty} g_t(x)e^{-ikx}dx$$

- The *Inverse Fourier Transform* of the density is given by:

$$g_t(x) = \hat{F}^{-1}(\hat{g}_t(k)) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \hat{g}_t(k)e^{ikx}dk$$

- We have that:

$$\hat{F}^{-1}(\hat{F}(g_t(x))) = g_t(x)$$

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Aside: Solving KFE With Fourier Transforms*

- The *Fourier transform* of the density is given by:

$$\hat{g}_t(k) := \hat{F}(g_t(x)) := \int_{-\infty}^{\infty} g_t(x) e^{-ikx} dx$$

- This transformation has the useful properties:

$$\hat{F}(\partial_t g_t(x)) = \int_{-\infty}^{\infty} \partial_t g_t(x) e^{-ikx} dx = \partial_t \int_{-\infty}^{\infty} g_t(x) e^{-ikx} dx = \partial_t \hat{g}_t(k)$$

$$\hat{F}(\partial_{xx} g_t(x)) = \int_{-\infty}^{\infty} \partial_{xx} g_t(x) e^{-ikx} dx \stackrel{IBP}{=} (ik)^2 \hat{g}_t(k) = -k^2 \hat{g}_t(k)$$

$$\hat{F}(g_0(x)) = \int_{-\infty}^{\infty} \partial_t \delta(x - x_0) e^{-ikx} dx = e^{-i2\pi k x_0}$$

- So, the Fourier transfrom of the KFE becomes the linear ODE:

$$\partial_t \hat{g}_t(k) = -\sigma^2 k^2 \hat{g}_t(k), \quad \hat{g}_t(0) = e^{-i2\pi k x_0}$$

Aside: Solving KFE With Fourier Transforms

- Solving the linear ODE, we get that:

$$\hat{g}_t(k) = e^{-i2\pi k x_0} e^{-\sigma^2 k^2 t}$$

- We can then finally invert the Fourier transform:

$$\begin{aligned} g_t(x) = \hat{F}^{-1}(\hat{g}_t(k)) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{g}_t(k) e^{ikx} dk \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i2\pi k x_0} e^{-\sigma^2 k^2 t} e^{ikx} dk \\ &= \frac{1}{\sqrt{4\pi\sigma^2 t}} e^{-\frac{(x-x_0)^2}{4\sigma^2 t}} \end{aligned}$$

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Brownian Motion in $\mathbf{R}$ (No Boundary)

- POPULATION: Continuum of agents with states, $X_t^i \in \mathbb{R}$, defined by:

$$dX_t^i = \sigma dW_t^i, \quad X_0^i = x_0$$

- KFE: The density, $g_t(x)$, satisfies the KFE:

$$\partial_t g_t(x) = \sigma^2 \partial_{xx} g_t(x), \quad s.t.$$
$$g_0(x) = \delta(x - x_0)$$

where δ is the Dirac-delta function (centered at x_0).

This is also known as **heat or diffusion equation**.

Aside: Recall the Dirac-Delta Function

- Informally, the Dirac-Delta function, $\delta(x - x_0)$ is a function satisfying:

$$\delta(x) = \begin{cases} 0, & \text{if } x \neq x_0 \\ \infty, & \text{if } x = x_0 \end{cases}$$
$$\int_{-\infty}^{\infty} \delta(x) = 1$$

- Conceptually, it is the density for a "point-mass" at x_0

Brownian Motion in $\mathbf{R}$ (No Boundary)

- Two methods for getting the population density:
  1. Solve the KFE (using Fourier transforms) Details
  2. Use normal distribution of $X_t^i = X_0^i + \sigma W_t^i \sim N(x_0, \sigma^2 t)$

- Population density: (all agents start at x_0)

$$g_t(x|x_0) = \frac{1}{\sqrt{4\pi\sigma^2 t}} e^{-\frac{(x-x_0)^2}{4\sigma^2 t}}$$

  - Function is known as Green's function or the fundamental solution,
  - No stationary distribution since $g_t(x) \to 0$, as $n \to \infty$.

- Population density: (from initial density $\psi(x)$): If the initial population density is $\psi(x_0)$, then the population density t is:

$$g_t(x) = \int_{-\infty}^{+\infty} g_t(x|x_0)\psi(x_0)dx_0$$

Brownian Motion With Boundary Conditions in the Appendix

- Absorbing boundaries Details

$$g_t(x|x_0) = 2\sum_{k=1}^{\infty} e^{-(k\pi\sigma)^2 t} \sin\left(k\pi x_0\right) \sin\left(k\pi x\right)$$

- Reflecting boundaries Details

$$g_t(x|x_0) = 1 + 2\sum_{k=1}^{\infty} e^{-(k\pi\sigma)^2 t} \cos\left(k\pi x_0\right) \cos\left(k\pi x\right)$$

Brownian Motion with Absorbing Boundaries

- Population: Continuum of agents with states, $X_t^i \in [0,1]$, defined by:

$$dX_t^i = \sigma dW_t^i, \quad X_0^i = x_0$$

  with aborbing boundaries at $\{0,1\}$.

- KFE: The density, $g_t(x)$, satisfies the KFE:

$$\begin{aligned} \partial_t g_t(x) &= \sigma^2 \partial_{xx} g_t(x), \quad s.t. \\ g_0(x) &= \delta(x - x_0) \\ g_t(0) &= g_t(1) = 0 \end{aligned}$$

  where δ is the Dirac-delta function. This type of boundary condition is called a **Dirichlet** boundary condition.

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Brownian Motion with Absorbing Boundaries

- Look for a solution as a **trigonometric Fourier series**:

$$g_t(x) = \sum_{k=1}^{\infty} \left(a_{k,t} \cos(k\pi x) + b_{k,t} \sin(k\pi x) \right)$$

where $a_{n,t}$ and $b_{n,t}$ are called the **Fourier coefficients** and are given by:

$$a_{n,t} = \int_0^1 g_t(x) \cos(k\pi x)\, dx, \quad b_{n,t} = \int_0^1 g_t(x) \sin(k\pi x)\, dx$$

- Population density: Imposing initial and boundary conditions gives:

$$g_t(x|x_0) = 2 \sum_{k=1}^{\infty} e^{-(k\pi\sigma)^2 t} \sin(k\pi x_0) \sin(k\pi x)$$

- Observe that:
  - BCs: set $a_{k,t} = 0$ because we need zero sum at $x = 1$,
  - No stationary distribution: $g_t(x|x_0) \to 0$, as $t \to \infty$

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Ornstein-Uhlenbeck Process

- Population: Suppose that the state variable of each agent, $X_t^i \in \mathbb{R}$, follows the **Ornstein-Uhlenbeck (OU)** process:

$$dX_t^i = -\alpha X_t^i dt + \sigma dW_t^i, \quad X_0^i = x_0 \tag{3}$$

  - This process is mean reverting around zero,
  - Can interpret the OU process as the continuous time version of the AR(1) process since the discrete approximation is that:

$$\Delta X_t^i = -\alpha X_t^i + \sigma \Delta W_t^i, \quad \Delta W_t^i = W_{t+1}^i - W_t^i \sim N(0,1)$$

- KFE: for the population density is given by:

$$\partial_t g_t(x) = \alpha \partial_x [x g_t(x)] + \frac{\sigma^2}{2} \partial_{xx} g_t(x), \quad g_0(x) = \delta(x - x_0) \tag{4}$$

Ornstein-Uhlenbeck Process

- Homework exercise: solution to (3) is given by:

$$X_t^i = e^{-\alpha t}\left(x_0 + \int_0^t \sigma e^{\alpha s} dW_s^i\right)$$

- Again, there are two solution methods for finding g_t:
  1. Solve the KFE, equation (4) (e.g. using Fourier transforms)
  2. Use distribution of X_t^i

- Population density: (all agents start at x_0):

$$g_t(x|x_0) = \sqrt{\frac{\alpha}{\pi\sigma^2(1-e^{-2\alpha t})}} \exp\left(-\frac{\alpha(x - x_0 e^{-\alpha t})^2}{\sigma^2(1-e^{-2\alpha t})}\right)$$

- Population density: (initial population density $\psi(x)$):

$$g_t(x|x_0) = \int_{-\infty}^{+\infty} g_t(x|x_0)\psi(x_0)dx_0$$

Ornstein-Uhlenbeck Process

- Property 1: Stationary distribution:

$$\lim_{t\to\infty} g_t(x|x_0) = \sqrt{\frac{\alpha}{\pi\sigma^2}} \exp\left(-\frac{\alpha x^2}{\sigma^2}\right)$$

- Property 2: The moments of the OU process are defined by:

$$M_k(t) := \mathbb{E}[(X_t)^k] = \int_{-\infty}^{\infty} x^k g_t(x) dx, \quad k = 0, 1, 2, \ldots$$

Exercise: show that the moments of the distribution are:

$$\begin{aligned}
M_0(t) &= M_0(0) = 1 \\
M_1(t) &= M_1(0) e^{-\alpha t} \\
M_k(t) &= e^{-\alpha k t} M_k(0) + \frac{1}{2}\sigma^2 k(k-1) \int_0^t e^{-\alpha k(t-s)} M_{k-2}(s) ds
\end{aligned}$$

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Geometric Brownian Motion

- Population: Suppose that the state variable of each agent, $X_t^i \in \mathbb{R}$, follows the **Geometric Brownian Motion** process:

$$dX_t^i = \mu X_t^i dt + \sigma X_t^i dW_t^i, \quad s.t. \quad X_0^i = X_0 \tag{5}$$

  - This process is a "stochastic" exponential growth process,

- KFE: for the population is given by:

$$\begin{aligned} \partial_t g_t(x) = & -\partial_x[\mu x g_t(x)] + \frac{1}{2}\partial_{xx}[\sigma^2 x^2 g_t(x)], s.t. \\ g_0(x) = & \ \delta(x - x_0) \end{aligned} \tag{6}$$

Geometric Brownian Motion

- SOLUTION: Solution to SDE (5) is given by:

$$X_t^i = x_0 e^{\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma W_t^i}$$

- Two methods for solving for g_t:
  1. Solve the KFE, equation (6) (using Fourier transforms)
  2. Use distribution of $X_t^i \sim \text{logN}\left(\log(x_0) + \left(\mu - \frac{1}{2}\sigma^2\right)t, \sigma^2 t\right)$

- POPULATION DENSITY: (all agents start at x_0):

$$g_t(x|x_0) = \frac{1}{x\sqrt{2\pi\sigma^2 t}} \exp\left(-\frac{(\log(x) - \log(x_0) - \left(\mu - \frac{1}{2}\sigma^2\right)t)^2}{2\sigma^2 t}\right)$$

- POPULATION DENSITY (initial population density $\psi(x)$):

$$g_t(x|x_0) = \int_{-\infty}^{+\infty} g_t(x|x_0)\psi(x_0)dx_0$$

Geometric Brownian Motion

- Property 1: No stationary distribution.

- Property 2: The moments of the GBM process are defined by:

$$M_k(t) := \mathbb{E}[(X_t)^k] = \int_{-\infty}^{\infty} x^k g_t(x) dx, \quad k = 0, 1, 2, \ldots$$

Can show that moments are:

$$M_1(t) = e^{\mu t} M_1(0)$$

$$M_k(t) = e^{\left(\mu + (k-1)\frac{1}{2}\sigma^2\right)kt} M_k(0)$$

  - $M_k(t)$ might diverge as $t \to \infty$, depending on the values of μ and σ,
  - E.g., for the second moment, we have that:

$$M_2(t) = e^{(2\mu + \sigma^2)t} M_2(0)$$

  which diverges when $2\mu + \sigma > 0$.

Brownian Motion with Reflecting Boundaries

- Population: Continuum of agents with states, $X_t^i \in [0,1]$, defined by:

$$dX_t^i = \sigma dW_t^i, \quad X_0^i = x_0$$

with reflecting boundaries at $\{0,1\}$.

- KFE: The density, $g_t(x)$, satisfies the KFE:

$$\begin{aligned}
\partial_t g_t(x) &= \sigma^2 \partial_{xx} g_t(x), \quad s.t. \\
g_0(x) &= \delta(x - x_0) \\
\partial_x g_t(0) &= \partial_x g_t(1) = 0
\end{aligned}$$

where δ is the Dirac-delta function. This type of boundary condition is called a **Neumann** boundary condition.

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Brownian Motion with Reflecting Boundaries

- Look for a solution as a **trigonometric Fourier series**:

$$g_t(x) = \sum_{k=1}^{\infty} \left(a_{k,t} \cos(k\pi x) + b_{k,t} \sin(k\pi x)\right)$$

where $a_{n,t}$ and $b_{n,t}$ are called the **Fourier coefficients** and are given by:

$$a_{n,t} = \int_0^1 g_t(x) \cos(k\pi x)\, dx, \quad b_{n,t} = \int_0^1 g_t(x) \sin(k\pi x)\, dx$$

- POPULATION DENSITY: Imposing initial and boundary conditions gives:

$$g_t(x|x_0) = 1 + 2\sum_{k=1}^{\infty} e^{-(k\pi\sigma)^2 t} \cos(k\pi x_0) \cos(k\pi x)$$

- Observe that:
  - BCs: set $b_{k,t} = 0$ because we need zero derivative at $x = 1$,
  - Stationary distribution: $g_t(x|x_0) \to 1$, as $t \to \infty$

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Stationary or Invariant Measures*

- Some examples had stationary distribution; others did not
- Now study conditions for when univariate processes on $\mathbb{R}$ have stationary distributions
- See Riskin (1996) for discussion of multivariate processes
- If a process has a unique stationary distribution, then it is **ergodic** (i.e. the invariant measure is the long run time average)

Definitions*

- The **scale density**, $s(x)$, is:

$$s(x) = \exp\left(-\int_{x_0}^{x} \frac{2\mu_s}{\sigma_s} ds\right)$$

- The **scale function**, $S(x)$, is:

$$S(x) = \int_{x_0}^{x} s(u)du = \int_{x_0}^{x} \exp\left(-\int_{x_0}^{u} \frac{2\mu_s}{\sigma_s} ds\right) du$$

- The **speed density**, $m(x)$, is:

$$m(x) = \frac{1}{\sigma^2(x)s(x)}$$

- The **speed measure**, M, is

$$M(x) = \int_{x_0}^{x} m(u)du = \int_{x_0}^{x} \frac{1}{\sigma^2(u)} \exp\left(\int_{x_0}^{u} \frac{2\mu_s}{\sigma_s} ds\right) du$$

Scale Density is the "Natural" Scale*

- Let $T_a = \inf\{t > 0 : X_t = a\}$ denote the first time X reaches a

- Let $X_0 = x$ and $a < x < b$. Then,

$$\mathbb{P}(T_b < T_a | X_0 = x) = \frac{S(x) - S(a)}{S(b) - S(a)}$$

- So the "scaled distance" $S(x)$ is proportional to the probability

- Observe that:

$$\mathbb{P}(T_y < \infty | X_0 = x) = \lim_{a \to -\infty} \frac{S(x) - S(a)}{S(y) - S(a)}, \quad a < x < y$$

$$\mathbb{P}(T_y < \infty | X_0 = x) = \lim_{b \to \infty} \frac{S(x) - S(b)}{S(y) - S(b)}, \quad y < x < b$$

Recurrence and Transience*

- A point, x, is called **recurrent** for diffusion X if the probability of the process returning to x infinitely often is one

- A point, x, is called **transient** for diffusion X if

$$\mathbb{P}\left(\lim_{t\to\infty} |X_t| = \infty | X_0 = x\right) = 1$$

Theorem

Let X be a diffusion with time invariance drift $\mu(X_t)$ and volatility $\sigma(X_t)$. Then, X is recurrent if and only if:

$$\lim_{b\to\infty} S(a) = \int_{x_0}^{b} \exp\left(-\int_{x_0}^{u} \frac{2\mu_s}{\sigma_s} ds\right) du = \infty, \quad \textit{and}$$

$$\lim_{a\to-\infty} S(b) = \int_{a}^{x_0} \exp\left(-\int_{x_0}^{u} \frac{2\mu_s}{\sigma_s} ds\right) du = \infty$$

Stationary Distribution*

Theorem

Let X be a diffusion with time invariance drift $\mu(X_t)$ and volatility $\sigma(X_t)$. Suppose that μ and σ are twice continuously differentiable with second derivatives satisfying a Holder condition. Then a stationary (or invariant) density exists if and only if:

1. *X is recurrent:*

$$\lim_{b\to\infty} S(a) = \int_{x_0}^{b} \exp\left(-\int_{x_0}^{u} \frac{2\mu_s}{\sigma_s} ds\right) du = \infty, \quad \textit{and}$$
$$\lim_{a\to-\infty} S(b) = \int_{a}^{x_0} \exp\left(-\int_{x_0}^{u} \frac{2\mu_s}{\sigma_s} ds\right) du = \infty$$

2. *The stationary density is integrable:*

$$M(x) = \int_{x_0}^{x} m(u)du = \int_{-\infty}^{+\infty} \frac{1}{\sigma^2(u)} \exp\left(\int_{x_0}^{u} \frac{2\mu_s}{\sigma_s} ds\right) du < \infty$$

Exercise*

- Exercise: verify that Geometric Brownian motion fails the conditions
- Exercise: verify that the Ornstein-Uhlenbeck process passes the conditions

Stationary Density

- If X has a stationary density, then the KFE becomes:

$$0 = -\partial_x[\mu(x)g(x)] + \frac{1}{2}\sigma_{xx}[\sigma^2(x)g(x)]$$

- Integrate once with respect to x gives (for constant C_1):

$$\frac{1}{2}C_1 = -\mu(x)g(x) + \frac{1}{2}\sigma_x[\sigma^2(x)g(x)]$$

(This equation says that the flux is constant across the domain)

- Solve the first order o.d.e. to get:

$$g(x) = \frac{1}{\sigma^2(x)} \exp\left(\int_{x_0}^{x} \frac{2\mu(\xi)}{\sigma^2(\xi)} d\xi\right) \left(C_1 \int_{x_0}^{x} \exp\left(-\int_{x_0}^{u} \frac{2\mu_s}{\sigma_s} ds\right) du + C_2\right)$$

where C_1 and C_2 are pinned down by the boundary conditions.